بسم الله الرحمن الرحيم

in the name of Allah, the most beneficient,
the most merciful

Al Ryada University for Science and Technology

RST

Eukaryotic cells , extracellular fluid and Water Balance

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By the end of the lecture the student will be able to:

1. Describe the biochemical structure of cells.
2. List the two main types of cells
3. Define apoptosis and neoplasia.
4. Enumerate cell membrane functions.
5. Define extra and intracellular fluids

- A **body system** is a **group of organs that work together** to perform a specific function necessary for survival.
- **Organs:** made of different tissues working together.
- **Tissues** – groups of similar cells performing a function.
- A **cell** is the **smallest unit of life** that can carry out all vital functions such as metabolism, growth, response to stimuli, and reproduction. All living organisms are made of one or more cells.

What Is the Chemical Structure of Cells?

Cells are composed of biomolecules:

1- Water: (≈70%)

2- Proteins (enzymes, structural components)

3- Lipids (membranes, energy storage)

4- Carbohydrates (energy, cell recognition)

5-Nucleic acids (DNA & RNA)

6-Hydrogen, oxygen, nitrogen, carbon, sulfur, and phosphorus normally makeup the main mass of living cells.

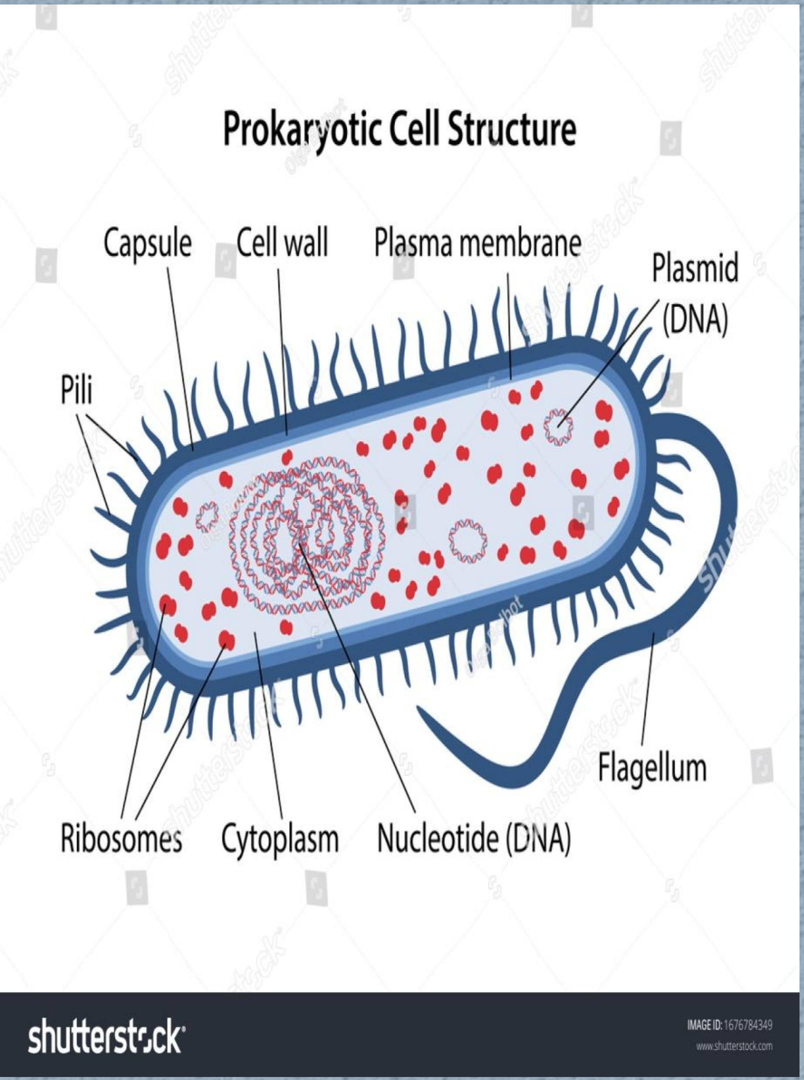
-These molecules build cellular structures and support life processes.

- The chemical complexity differs significantly between prokaryotes and eukaryotes.

-Most of diseases in animals, humans are manifestations of abnormalit1es in biomolecules, chemical reactions, or biochemical pathways

- Prokaryotes - Small, single cell organisms lacking a membrane-bound nucleus and other subcellular compartments(organelles). Bacteria are examples.

- Eukaryotes - Cell or organism with membrane-bound nucleus and other well-developed subcellular compartments (organelles). Yeast, fungi, plants, animals and humans are eukaryotes.


Prokaryotic Cell Structure
Capsule
Cell wall
Plasma membrane
Plasmid
(DNA)
Pili
Flagellum
Ribosomes
Cytoplasm
Nucleotide (DNA)
shutterstock
IMAGE ID: 1676784349
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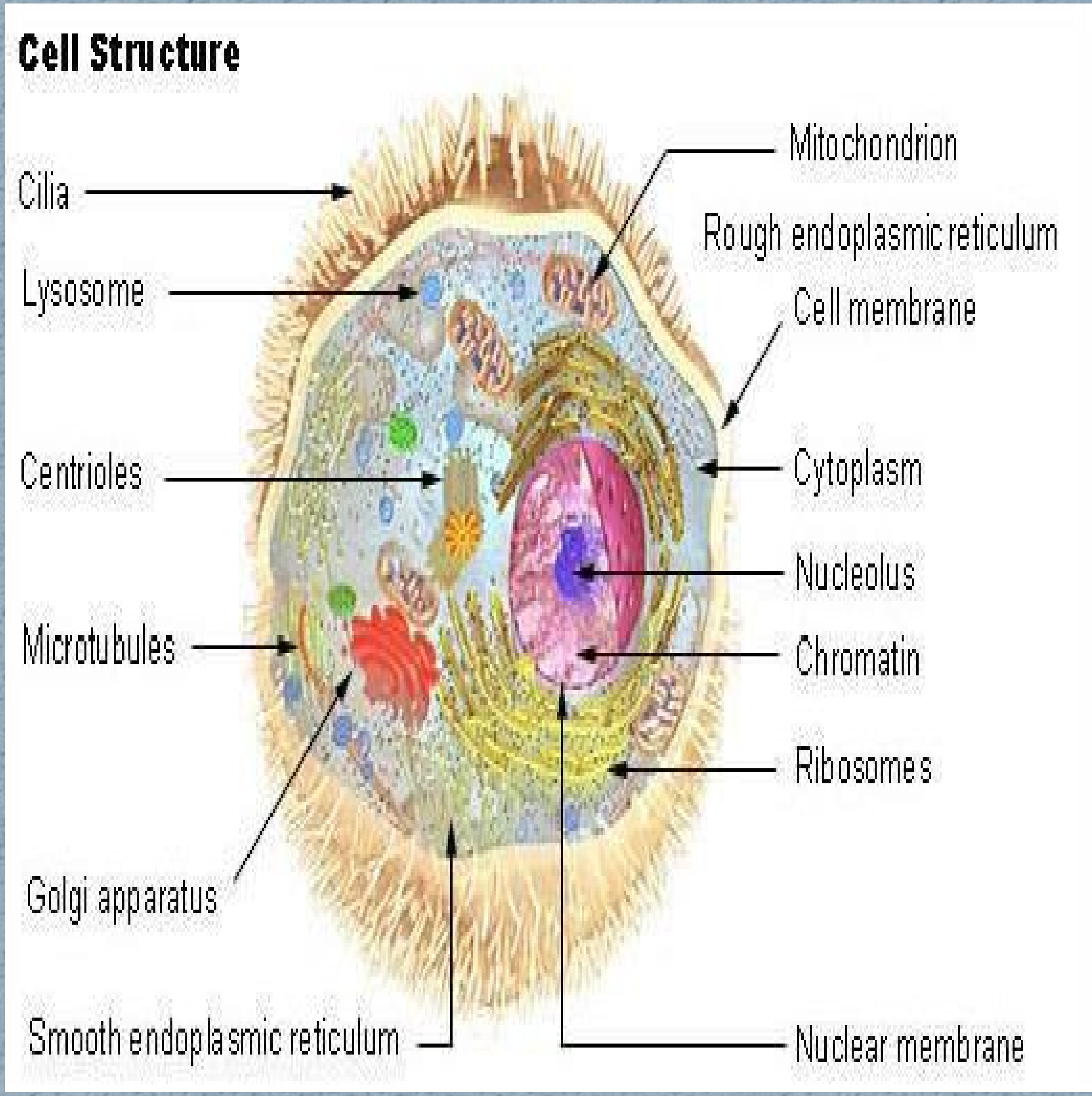

Cell Structure
Cilia
Mitochondrion
Rough endoplasmic reticulum
Lysosome
Cell membrane
Centrioles
Cytoplasm
Nucleolus
Microtubules
Chromatin
Ribosomes
Golgi apparatus
Smooth endoplasmic reticulum
Nuclear membrane

Prokaryotic Cells – Chemical Features

- Simple internal chemical organization
- No membrane-bound nucleus
- DNA is circular and located in nucleoid region
- Cell membrane made of phospholipids(No sterols)
- Cell wall contains peptidoglycan (bacteria)
- Ribosomes are smaller (70S)

Eukaryotic Cells – Chemical Features

- Highly organized chemical environment
- True nucleus with nuclear membrane
- Linear DNA associated with histone proteins
- Membrane-bound organelles (mitochondria, ER, Golgi)
- Cell membrane contains phospholipids and sterols
- Ribosomes are larger (80S)

Chemical Structure: Cell Membrane

Prokaryotic membrane:
- Phospholipid bilayer
- No sterols (except mycoplasma)

Eukaryotic membrane :
- Phospholipid bilayer with cholesterol
- More complex membrane proteins

Genetic Material Comparison

Prokaryotes:

- Circular DNA
- No histones
- Plasmid

Eukaryotes:

- Linear DNA
- DNA wrapped around histones forming chromatin
- No Plasmid

Organelles and Chemical Complexity

Prokaryotes:

- No membrane-bound organelles

Eukaryotes:

- Organelles with specialized chemical functions:
  * Mitochondria – ATP synthesis
  * Lysosomes – digestion
  * Endoplasmic Reticulum(ER) – protein synthesis

Summary Table

Prokaryotes:

- Simpler chemical organization
- Smaller size

Eukaryotes:

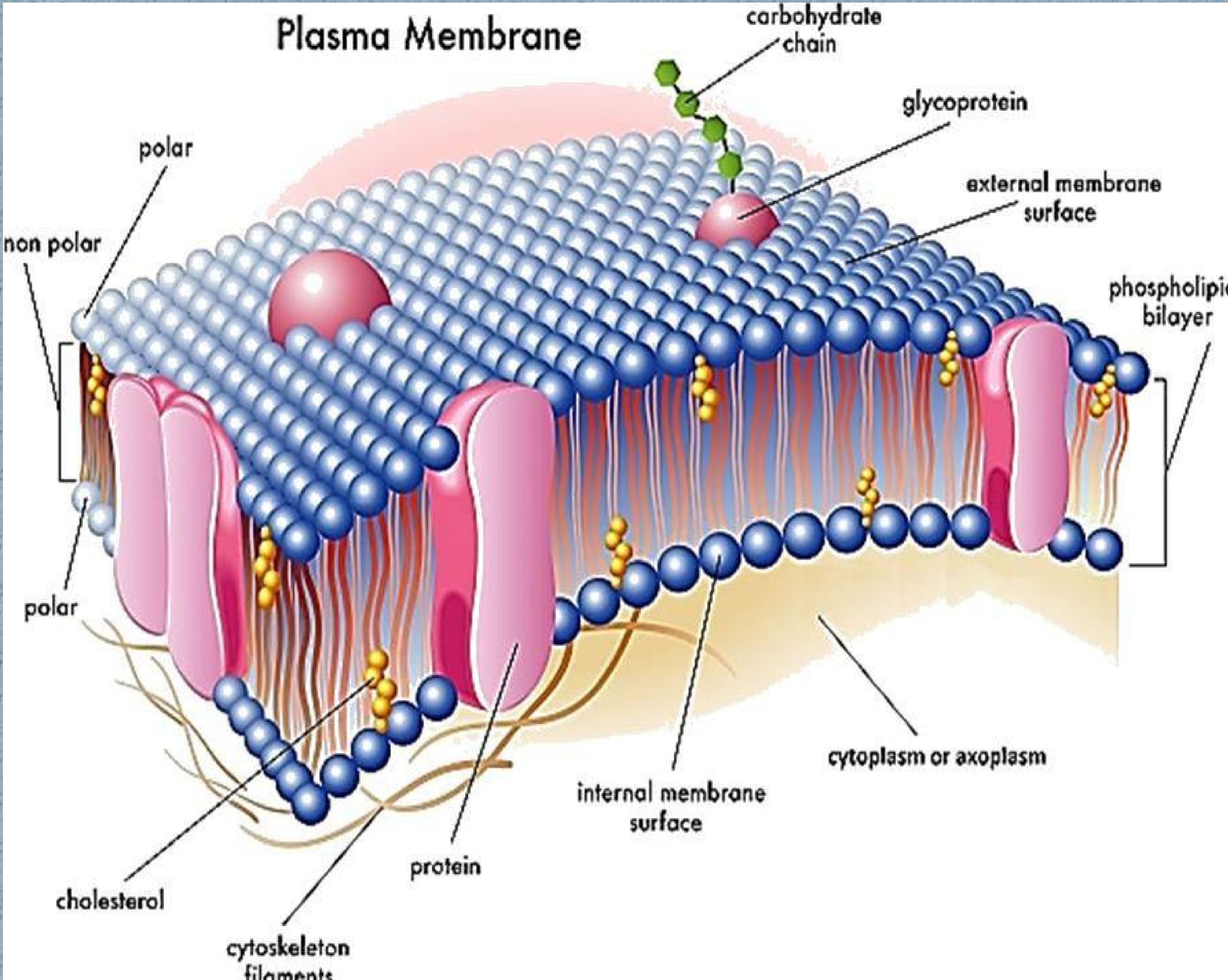
- More complex chemistry
- Larger and highly compartmentalized

Comparison of Chemical Structures

Feature	Prokaryotic Cells	Eukaryotic Cells
Genetic Material	Circular DNA, no nucleus	Linear DNA, nucleus
DNA Packaging	No histones	Histone-associated chromatin
Cell Membrane	Phospholipids only	Phospholipids + sterols
Organelles	Absent	Present
Ribosomes	70S	80S

Definition of Apoptosis: It is a programmed cell death. It is a controlled process of cellular self-destruction that requires certain protein releases and specific cellular signals

Neoplasia: is an abnormal accumulation of cells that occurs because of an imbalance between cellular proliferation and cellular attrition.


Plasma Membrane
carbohydrate chain
glycoprotein
external membrane surface
phospholipid bilayer
polar
non polar
polar
cytoplasm or axoplasm
internal membrane surface
protein
cholesterol
cytoskeleton filaments

Membranes are thin envelopes that define the volume of cells and define cellular organelles. They have many essential cellular functions:

1Membranes constitute the boundaries of cells and intracellular organelles. They provide a surface where many important biological reactions, processes occur and interaction between cells.

2 Membranes have proteins that mediate and regulate the transport of metabolites, macromolecules and ions.

3Hormones and many other biological signal molecules and regulatory agents exert their effects via interaction with membranes which have specific receptors.

4-Electron transport, oxidative phosphorylation, muscle contraction and electrical activity all depend on membranes and membrane proteins.

5-Plasma membranes exchange material with extracellular environment by exocytosis and endocytosis.

6- Intracellular membranes form many of the morphologically distinguishable organelles **e.g** mitochondria, Golgi complex, endoplasmic reticulum, lysosomes and nuclear membranes.

Water Balance

WATER accounts for about 50 – 60% of an average person's weight.

- Fat tissue has lower percentage of water and women tend to have more fat, so the percentage of water in the average woman is lower (**50 to 55%**) than it is in the average man (**60%**).
- The percentage of water is also lower in **older** and **obese people**.

- A 70 kilogram man has about 45 liters of water in
- his body:

1. **Intracellular Fluid** (2/3 of body water) = 30 L
2. **Extracellular Fluid** (1/3 of body water) = 15 L

- Interstitial Fluid (4/5 of extracellular fluid)= 12 L
- Plasma (1/5 of extracellular fluid) = 3 L

How the body can regulates water?

1. The body obtains water primarily by **absorbing it from the digestive tract** in addition to the **small amount of water produced during metabolism.**

2. Water intake must balance water loss to maintain water balance and -- to protect against dehydration, the development of kidney stones, and other medical problems .

3. Healthy adults should drink at least 1.5 - 2.0 liters/day.

4. Drinking too much is better than drinking too little, because excreting excess water is much easier for the body than conserving water.

5. **Antidiuretic hormone (ADH):** secreted from the posterior lobe of pituitary gland maintain water at equilibrium through regulation of water absorption from the distal convoluted tubules of the kidney according to body needs.

6. The body regulates the amount of water in each of intracellular and extracellular compartments by **moving water between these areas** to keep1 **(Osmolality**) the amount in each area relatively constant .

Reference

1 Harpers illustrated Biochemistry

2 Lippincott illustrated reviews integrated systems


Thank you!